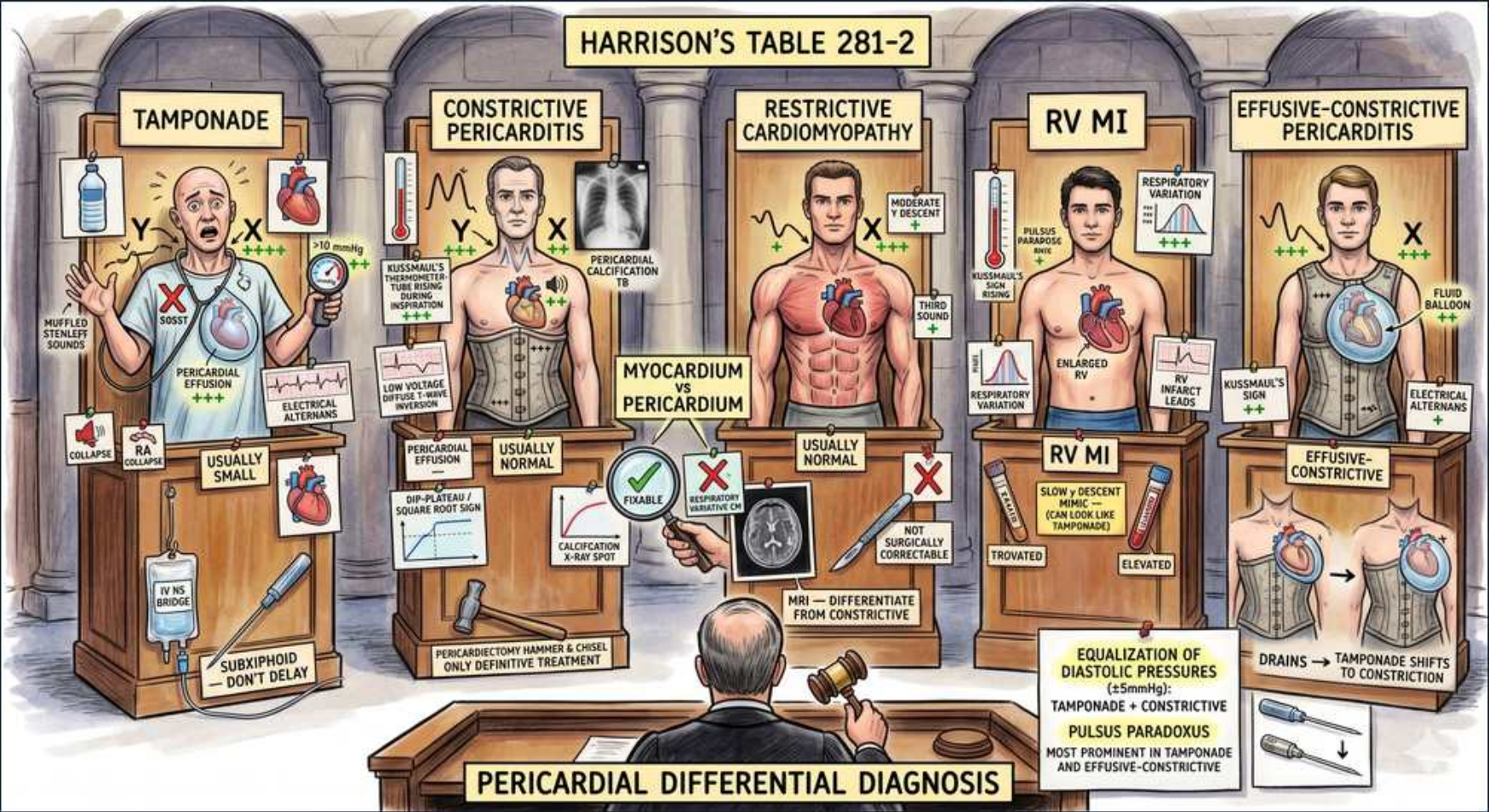


Pericardial Disease — Constrictive vs Restrictive (Harrison's 281-2)



1. Differential table

WHAT YOU SEE

A museum-style banner reading 'HARRISON'S TABLE 281-2' over five exhibit booths — the master differential table laid out booth by booth.

WHAT IT ENCODES

Master differential of conditions causing right-heart failure physiology with clear lungs: tamponade, constrictive pericarditis, restrictive cardiomyopathy, RV MI, and effusive-constrictive disease. All five share elevated JVP and near-equalized diastolic pressures; the discriminators are the respiratory variation pattern, the JVP descents (x vs y), imaging (pericardial thickness/calcification), BNP, and biopsy. Board strategy: anchor on whether the problem is the PERICARDIUM (fixable surgically) or the MYOCARDIUM (not surgically fixable).

2. Tamponade booth

WHAT YOU SEE

Leftmost booth 'TAMPONADE', muffled sounds, electrical alternans

WHAT IT ENCODES

Beck's triad = hypotension + muffled heart sounds + distended neck veins (JVP). Pulsus paradoxus >10 mmHg drop in systolic BP on inspiration is the key bedside sign; ECG shows low voltage and electrical alternans. Echo shows RA systolic collapse (sensitive) and RV early-diastolic collapse (specific) plus IVC plethora. JVP shows a blunted/absent y descent (X present, Y absent) because the chamber cannot fill in diastole against the surrounding fluid.

BOARD TRAP

WRONG to wait for a confirmatory CT or to give a large fluid bolus as definitive therapy. Tamponade is a clinical/echo diagnosis; the discriminator from constriction is that tamponade has an ABSENT y descent (only x), whereas constriction has a prominent y descent (rapid). Definitive treatment is pericardiocentesis/drainage, not imaging.

3. Tamponade = small/fast

WHAT YOU SEE

'USUALLY SMALL' note under tamponade

WHAT IT ENCODES

Rate of accumulation matters more than absolute volume: a rapidly accumulating effusion of only 150–200 mL (e.g., trauma, dissection, post-procedure) can tamponade because the stiff pericardium cannot stretch acutely. Drain via subxiphoid pericardiocentesis (echo-guided) without delay; surgical window if recurrent or loculated.

BOARD TRAP

WRONG to assume a small effusion 'can't be tamponade' — acute small effusions tamponade most readily. The pressure-volume curve of the pericardium is the discriminator: acute = steep curve (small volume, big pressure); chronic = shifted right (large volume tolerated).

4. Constrictive booth

WHAT YOU SEE

'CONSTRICTIVE PERICARDITIS' booth, Kussmaul, calcification

WHAT IT ENCODES

A rigid, scarred pericardium prevents diastolic filling. Hallmarks: Kussmaul sign (JVP fails to fall, or rises, with inspiration), pericardial knock (early high-pitched S3-like sound from abrupt cessation of filling), prominent rapid x AND y descents, and a 'dip-and-plateau'/square-root sign on ventricular pressure tracing. Catheterization shows DISCORDANT LV/RV systolic pressures with respiration (ventricular interdependence: inspiration → LV pressure falls, RV rises). BNP is normal or only mildly elevated. Common causes: prior viral/idiopathic pericarditis, cardiac surgery, mediastinal radiation, TB (worldwide).

BOARD TRAP

WRONG to label this restrictive cardiomyopathy. The discriminator is respiratory VARIATION/interdependence: constriction = DISCORDANT ventricular pressures + respiratory variation in mitral/tricuspid inflow (>25% mitral E variation), thickened/calcified pericardium on imaging, and NORMAL BNP. Treatment is pericardiectomy, not heart-failure medical therapy.

5. Pericardial calcification

WHAT YOU SEE

'PERICARDIAL CALCIFICATION' image in constrictive booth

WHAT IT ENCODES

Pericardial thickening >3–4 mm (best seen on cardiac CT/MRI) or eggshell calcification on lateral CXR strongly favors constriction over restrictive cardiomyopathy. Calcification is classic for chronic TB or remote viral pericarditis. Note: ~20% of constriction has a normal-thickness pericardium, so absence of thickening does not exclude it — use hemodynamics/MRI tagging if doubt persists.

BOARD TRAP

WRONG to require calcification before diagnosing constriction. Calcification is specific but insensitive; a substantial minority of surgically proven constriction shows no calcification. The presence of calcification rules IN pericardial disease, but its absence does not rule it out.

6. Pericardiectomy = only definitive

WHAT YOU SEE

'PERICARDIECTOMY... ONLY DEFINITIVE TREATMENT'

WHAT IT ENCODES

Complete pericardiectomy (surgical stripping of the rigid pericardium) is the only definitive cure for chronic constriction; diuretics give only temporary symptom relief. Operative mortality is significant (≈6–12%), worst with radiation-induced constriction (because the myocardium is often co-involved). Transient/early post-viral constriction may resolve with 2–3 months of anti-inflammatory therapy before committing to surgery.

BOARD TRAP

WRONG to offer pericardiectomy for restrictive cardiomyopathy — stripping a normal pericardium off a stiff myocardium gives no benefit. The discriminator before sending to surgery: confirm the pericardium (not the myocardium) is the constraining structure via CT/MRI thickness and discordant cath hemodynamics.

7. Restrictive CM booth

WHAT YOU SEE

'RESTRICTIVE CARDIOMYOPATHY' booth

WHAT IT ENCODES

A stiff, infiltrated MYOCARDIUM impairs diastolic filling while the pericardium is normal. Causes: amyloidosis (most common; look for low-voltage ECG with pseudo-infarct Q waves but THICK walls on echo — the 'voltage-mass mismatch'), sarcoidosis, hemochromatosis, endomyocardial fibrosis, Loeffler endocarditis. BNP is HIGH (myocardium is diseased), filling pressures show CONCORDANT LV/RV changes with respiration, and septal motion lacks the constrictive 'bounce'. Endomyocardial biopsy confirms the infiltrate (e.g., Congo-red apple-green birefringence in amyloid).

BOARD TRAP

Restrictive CM is NOT cured by pericardiectomy. WRONG to send these patients to surgery. Discriminators from constriction: HIGH BNP (vs normal in constriction), CONCORDANT ventricular pressures with respiration (vs discordant), tissue Doppler e' REDUCED/septal e' <8 (vs preserved or 'annulus paradoxus' increased in constriction), and a normal-thickness pericardium on MRI. Treat the underlying infiltrative disease.

8. Myocardium vs pericardium

WHAT YOU SEE

Central 'MYOCARDIUM or PERICARDIUM' with green check

WHAT IT ENCODES

The single most important board distinction: constriction = diseased PERICARDIUM (surrounding shell is the problem; myocardium is intrinsically normal → surgically fixable). Restrictive CM = diseased MYOCARDIUM (the muscle itself is stiff/infiltrated → not surgically fixable). Mnemonic: 'Constriction Constrains from outside; Restriction is Rigid muscle inside.' Both produce a prominent y descent and a square-root sign, so those shared features cannot tell them apart — use BNP, respiratory concordance, and imaging.

BOARD TRAP

WRONG to use the square-root sign or prominent x/y descents to separate the two — BOTH share these. The actual discriminators are respiratory ventricular interdependence (discordant = constriction), BNP (normal = constriction, high = restrictive), tissue Doppler, and pericardial imaging.

9. MRI differentiates

WHAT YOU SEE

'MRI — to differentiate from constrictive'

WHAT IT ENCODES

Cardiac MRI/CT measures pericardial thickness (>3–4 mm favors constriction) and shows calcification; real-time cine MRI demonstrates the septal bounce and ventricular interdependence. MRI also characterizes myocardial infiltration (late gadolinium enhancement in amyloid/sarcoid). If imaging and echo remain equivocal, simultaneous LV/RV catheterization (looking for discordance) or endomyocardial biopsy settles it.

BOARD TRAP

WRONG to rely on echo alone when constriction vs restriction is unclear — echo can be ambiguous. The discriminating tools are MRI/CT pericardial thickness, simultaneous LV/RV cath showing discordance (constriction) vs concordance (restriction), and biopsy for infiltrative myocardial disease.

10. RV MI booth

WHAT YOU SEE

'RV MI' booth, elevated troponin

WHAT IT ENCODES

RV infarction (usually with inferior STEMI from proximal RCA occlusion) mimics tamponade/constriction: elevated JVP, clear lungs, hypotension, Kussmaul sign, and a slow y descent. The discriminators are ELEVATED TROPONIN, ST elevation in right-sided lead V4R (most sensitive), and inferior ST elevation. Management is preload-DEPENDENT: give IV fluids, AVOID nitrates/diuretics/morphine (which drop preload and cause profound hypotension), and reperfuse.

BOARD TRAP

WRONG to give nitrates or diuretics for the hypotension/elevated JVP — RV MI is preload-dependent, so these precipitate shock. The discriminator from pericardial constriction is the elevated troponin plus V4R ST elevation; treat with fluids and revascularization, not pericardiectomy.

11. Effusive-constrictive booth

WHAT YOU SEE

'EFFUSIVE-CONSTRICTIVE' booth, fluid balloon

WHAT IT ENCODES

A hybrid: a pericardial effusion (tamponade-like) overlying a constricting visceral pericardium. The clue is that after pericardiocentesis removes the fluid and intrapericardial pressure falls to zero, the RIGHT ATRIAL pressure stays ELEVATED — unmasking the underlying constriction. Often TB or radiation/neoplastic etiology. Pulsus paradoxus is prominent before drainage.

BOARD TRAP

WRONG to assume drainage will normalize hemodynamics. The discriminator: persistently elevated RA/diastolic pressure after the effusion is removed signals effusive-constrictive disease, which may ultimately need visceral pericardiectomy rather than drainage alone.

12. Equalization of pressures

WHAT YOU SEE

'EQUALIZATION OF DIASTOLIC PRESSURES' note

WHAT IT ENCODES

Near-equalization of end-diastolic pressures across all four chambers (RA ≈ RV ≈ PA diastolic ≈ PCWP, typically within 5 mmHg) is seen in tamponade and constriction because an external constraint dominates filling. In restrictive CM the LV diastolic pressure usually exceeds RV by >5 mmHg, helping separate it.

BOARD TRAP

WRONG to read equalized diastolic pressures as proof of constriction specifically — tamponade also equalizes pressures. Use the respiratory pattern (discordance) and the y-descent morphology to distinguish; and remember restrictive CM typically does NOT fully equalize (LV>RV by >5 mmHg).

13. Pulsus paradoxus pattern

WHAT YOU SEE

'PULSUS PARADOXUS — most prominent in tamponade & effusive-constrictive'

WHAT IT ENCODES

Pulsus paradoxus = inspiratory fall in systolic BP >10 mmHg, from exaggerated ventricular interdependence within a fixed total cardiac volume. Most prominent in tamponade and effusive-constrictive disease; often absent or mild in pure (non-effusive) constriction, where Kussmaul sign predominates instead.

BOARD TRAP

WRONG to exclude tamponade because pulsus is absent — it can be absent in low-pressure tamponade (hypovolemia), severe LV dysfunction, ASD, or aortic regurgitation. Conversely, pulsus also occurs in severe asthma/COPD and massive PE, so it is not specific to pericardial disease.

14. Drains shift to constriction

WHAT YOU SEE

'DRAINS — tamponade shifts to constriction'

WHAT IT ENCODES

In effusive-constrictive disease, draining the effusion converts the hemodynamic picture from a tamponade pattern to a pure constrictive pattern (RA pressure stays high, y descent becomes prominent). This is the diagnostic 'reveal' and identifies patients who may need pericardiectomy of the visceral layer.

BOARD TRAP

WRONG to discharge the patient as 'tamponade resolved' once fluid is drained. The discriminator is the persistent elevation of RA pressure and emergence of a rapid y descent after drainage — that unmasked constriction drives ongoing symptoms and may require surgery.
